

TEAM 1 A

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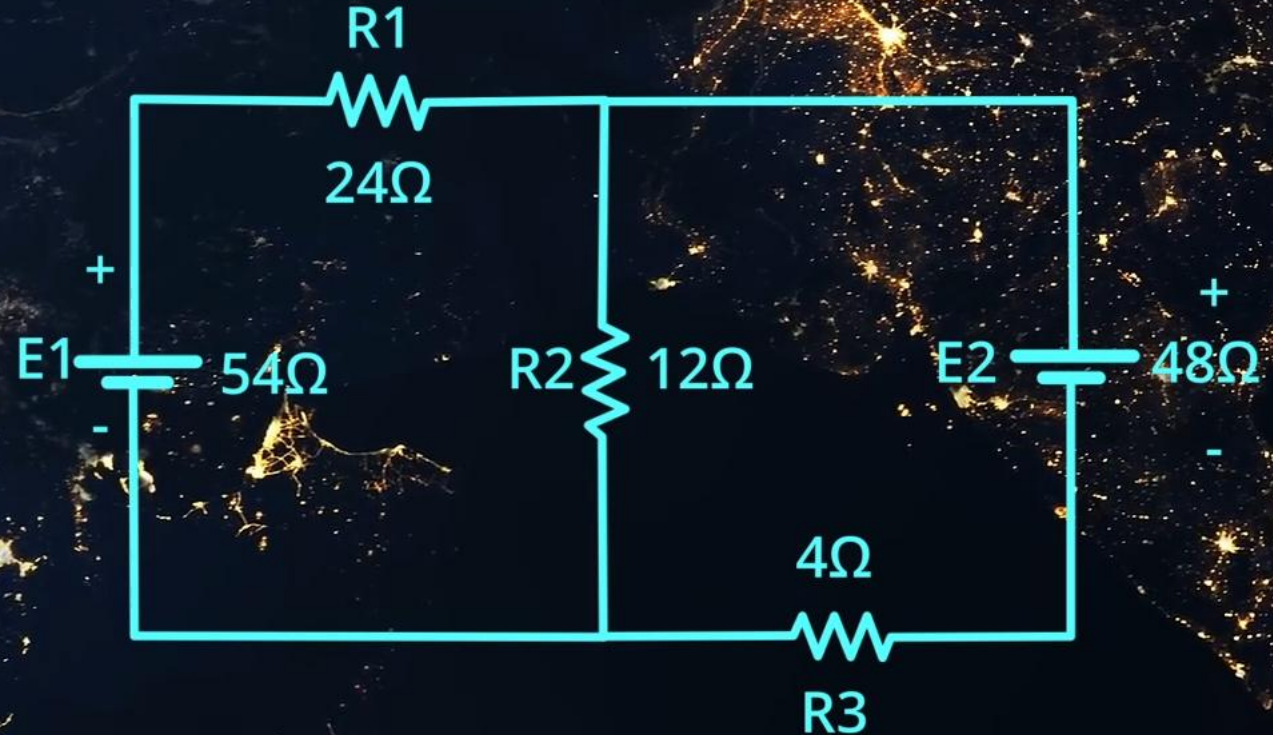
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QUESTION



Find the current through 4Ω resistor using super position theorem

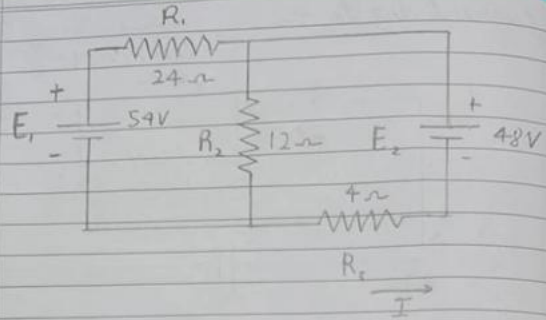
⚡ ANALYTICAL (MANUAL) SOLUTION

Circuit simplification
& setup



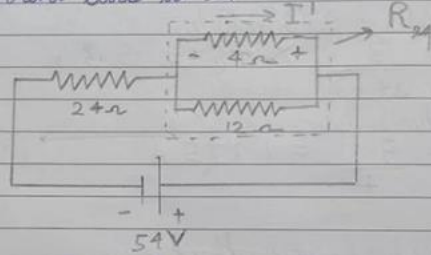
Applying laws &
final calculation

1)
a)



Find the current through 4Ω resistor using superposition theorem

Ans Current due to 54V



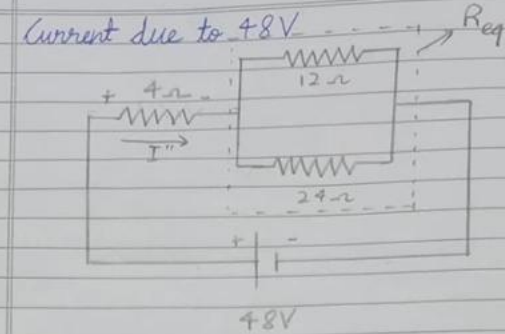
$$R_{eq} = \frac{1}{\frac{1}{4} + \frac{1}{12}} = 3\Omega$$

By Voltage Divider Rule,

$$V = \frac{3}{2+3} \times 54 = 6V$$

By Ohm's Law,

$$I' = -\frac{6}{4} = -1.5A$$



$$R_{eq} = \frac{1}{\frac{1}{12} + \frac{1}{24}} = 8\Omega$$

By Voltage Divider Rule,

$$V'' = \frac{4}{4+8} \times 48 = 16V$$

By Ohm's Law,

$$I'' = \frac{16}{4} = 4A$$

By Superposition Theorem,

$$\begin{aligned} I &= I' + I'' \\ &= -1.5 + 4 \\ &= 2.5A \end{aligned}$$

SUPER POSITION THEOREM

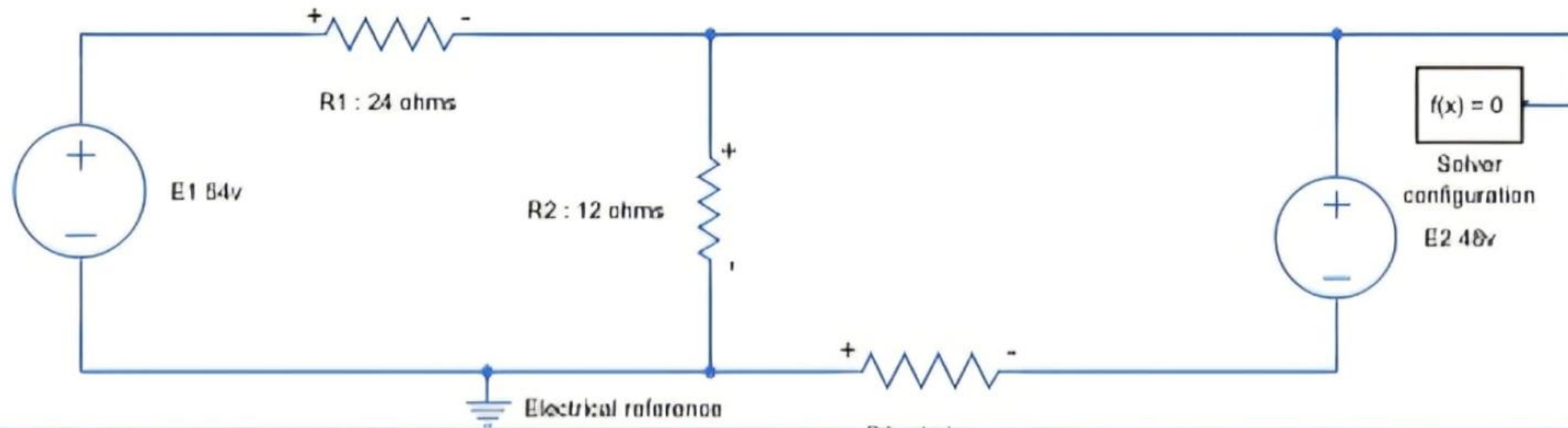
Superposition Theorem states that in any linear circuit with multiple independent sources, the total current or voltage in any element is the algebraic sum of the currents or voltages produced by each source acting alone

Replace other voltage sources with a short circuit.
Replace other current sources with an open circuit.

⚡ MATLAB SIMULATION

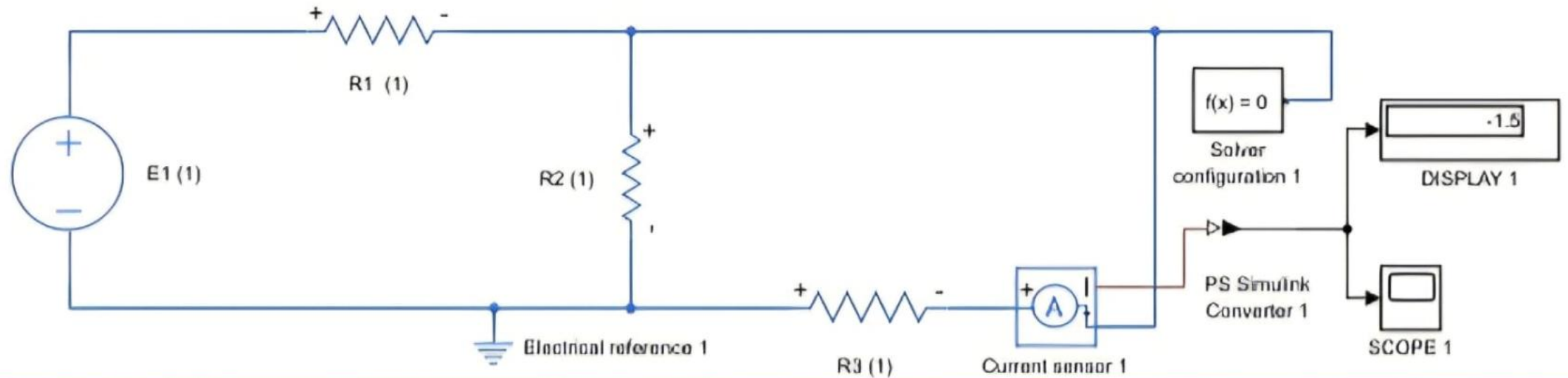
Circuit verification using simulation environment

Question



⚡ MATLAB SIMULATION

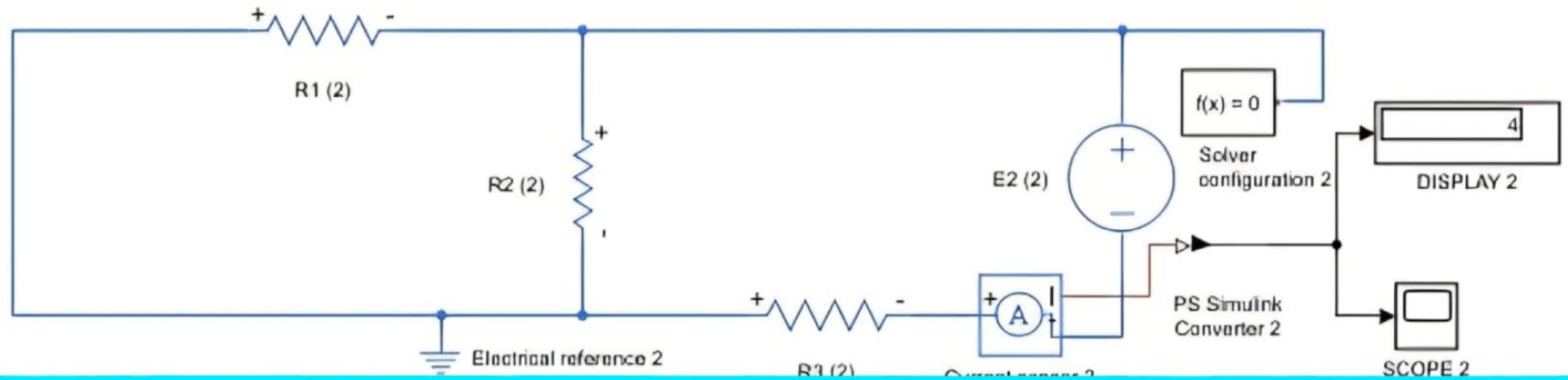
DUE TO 54V



⚡ MATLAB SIMULATION

DUE TO 48V

Due to 48v



⚡ MATLAB SIMULATION

DUE TO 54V AND 48V

Due to 54v and 48v

